MAGNATE ACADEMY

TECHNICAL ROUND 1 – DATA SCIENCE

Candidate Name:
Date: **Total Marks: 50**

Instructions

- Answer all questions.
- Write Python code wherever requested.
- Give clear explanations and examples where appropriate.
- The practical task will be evaluated separately as a complete Data Science workflow.

PART 1 – PYTHON | 20 MARKS

A. Python Coding | 10 Marks

1. Fibonacci Sequence — 2 Marks

Write a Python program to generate the Fibonacci sequence up to **n terms**.

2. String Manipulation — 2 Marks

Write a Python program to capitalise the **first and last character of every word** in a given string.

Example:
`hello a world` → `HellO A WorlD`

3. Loop with `else` — 2 Marks

Write a Python program using a **`for` or `while` loop with `else`**, without using an `if` statement.

4. Prime Number — 2 Marks

Write a Python program to determine whether a given number is **prime or not**.

5. Longest Word — 2 Marks

Write a Python program to find the **longest word in a given string**.

B. Python Theory | 10 Marks

1. What is Python?. — 1 Mark

2. What is a Lambda Function? Give an example. — 1 Mark

3. Differentiate between List, Tuple, Set and Dictionary. — 1 Mark

4. What is List Comprehension? Give an example. — 1 Mark

5. What are Generators? Why are they useful? — 1 Mark

6. What are Decorators in Python? — 1 Mark

7. What is a Function? Why do we use functions? — 1 Mark

8. Differentiate between `for` and `while` loops. — 1 Mark

9. What is Recursion? Give a suitable example. — 1 Mark

10. Differentiate between `break`, `continue`, and `pass`. — 1 Mark

PART 2 – MACHINE LEARNING | 15 MARKS

1. What is Machine Learning? Explain its major types. — 2 Marks

2. Differentiate between Classification and Regression with examples. — 2 Marks

3. Differentiate between a Loss Function and a Cost Function. — 2 Marks

4. What are the key assumptions of Linear Regression? — 2 Marks

5. What is Ridge Regression (L2 Regularisation)? Why is it used? — 2 Marks

6. What is Gradient Descent? Explain how it works. — 2 Marks

7. What is Ensemble Learning? Why does combining multiple models often improve performance? — 3 Marks

PART 3 – MACHINE LEARNING PRACTICAL | 15 MARKS

Complete Data Science Project

Task

Download a **new and suitable dataset** and build a complete Machine Learning project.

Your project must include:

Requirement	Marks
Dataset selection and loading	2
Data preprocessing	3
Exploratory Data Analysis and visualisation	2
Feature/target selection and train-test split	2
Model selection and training	2
Model evaluation and interpretation	2
GitHub repository and project documentation	2
Total	**15**

GitHub Requirement

Create a GitHub repository named : `Technical round 1`

Push the complete project to the repository, including the **source code, preprocessing, visualisations, model training, evaluation and `README.md`**.

> **Note: All stages must be completed in the given order. If any stage is skipped, the marks for that stage and all subsequent stages will be zero, even if the subsequent stages are completed. For example, skipping Stage 3 results in 0 marks for Stages 3–7.**